

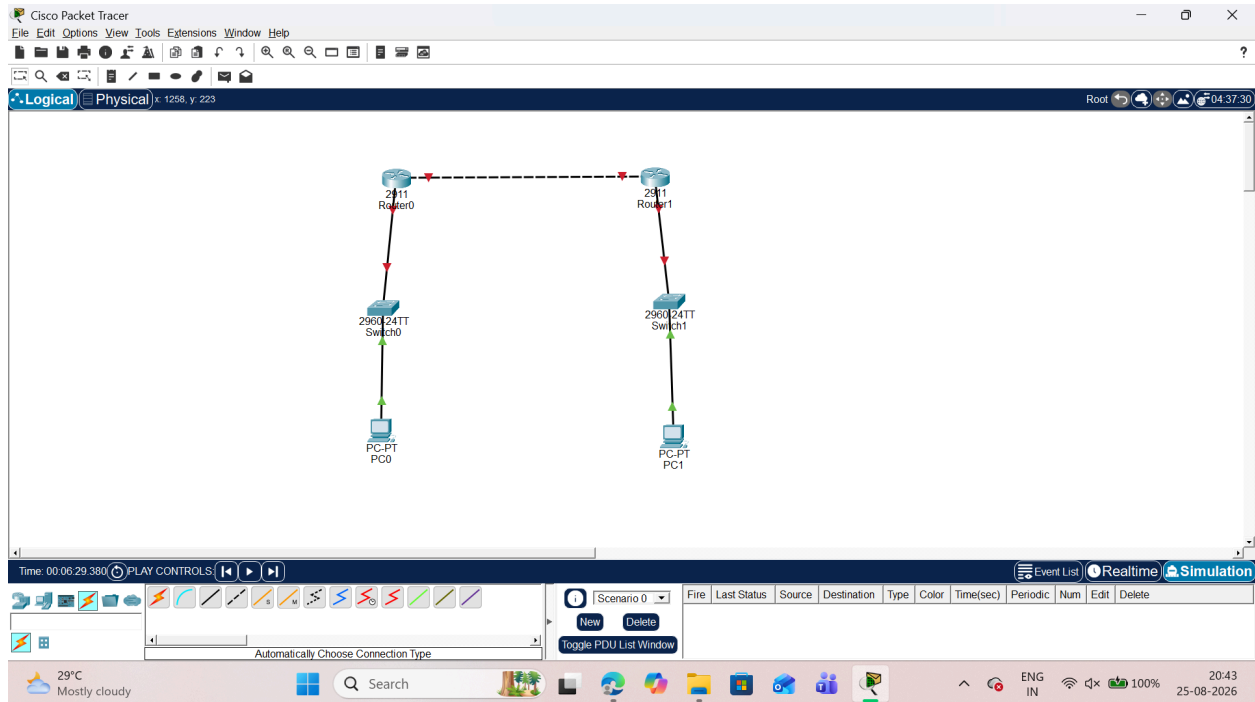
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SUBJECT: CYBER & INFORMATION SECURITY

PRACTICAL - 6

AIM:IP Security (IPsec) Configuration

Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

Step 1. Network topology



Step 2. Configure PC IP addresses a.PC0

The screenshot shows the configuration window for PC0. The 'IP Configuration' tab is selected, displaying the following settings for the FastEthernet0 interface:

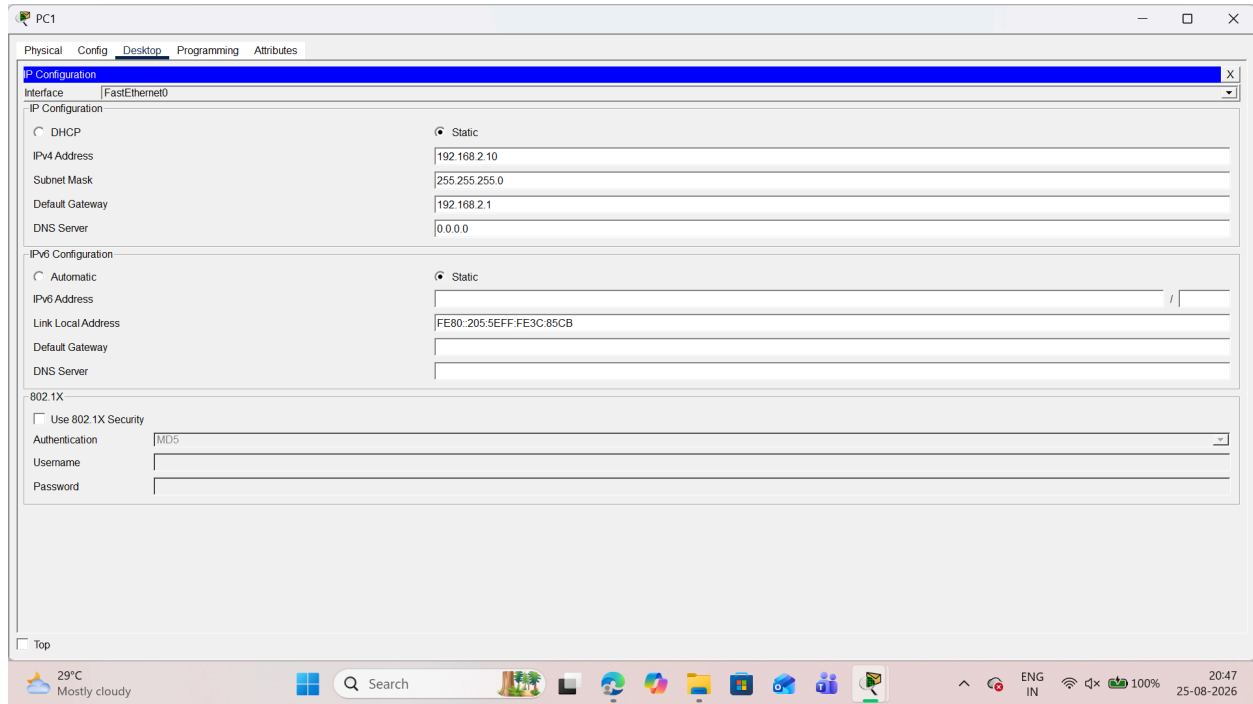
- IP Configuration:**
 - Interface: FastEthernet0
 - IP Configuration: Static
 - IPv4 Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - IPv6 Configuration: Static
 - IPv6 Address: FE80::201:42FF:FE39:CB3D
 - Link Local Address: FE80::201:42FF:FE39:CB3D
 - Default Gateway:
 - DNS Server:
- 802.1X:**
 - Use 802.1X Security:
 - Authentication: MD5
 - Username:
 - Password:

The bottom status bar shows the time is 20:44 and the date is 25-08-2026.

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T125

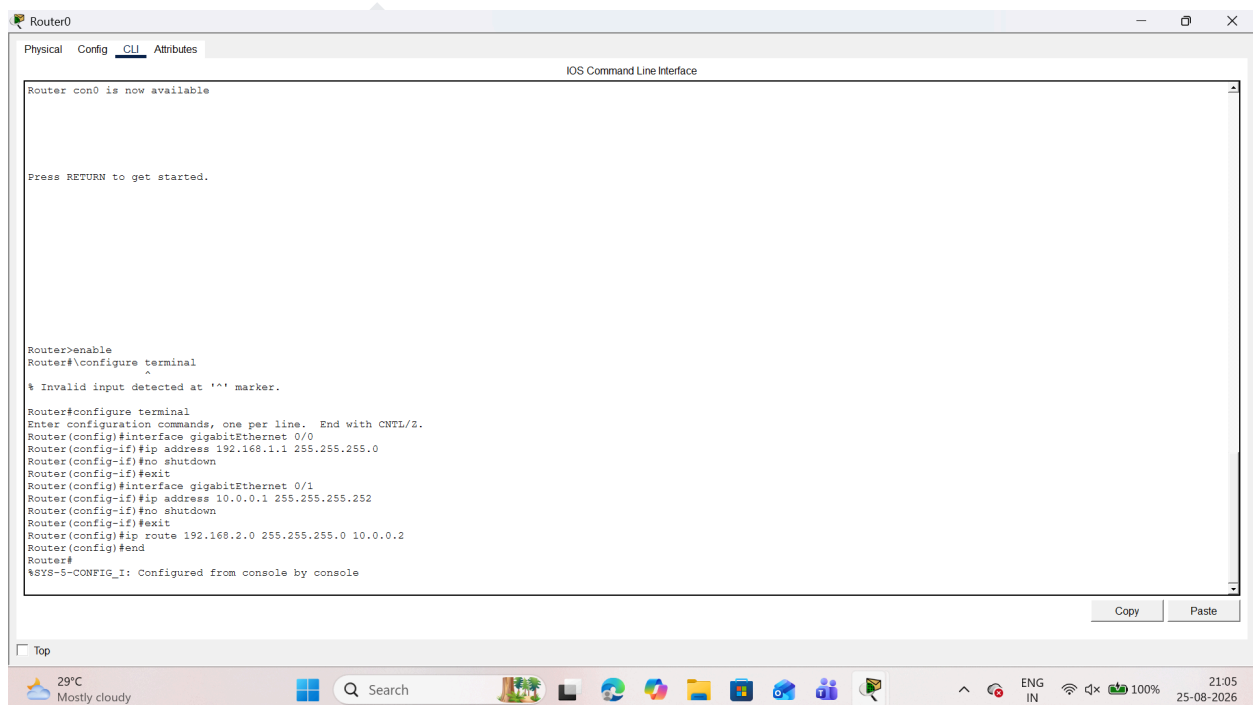
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b.PC1



Step 3. Configuring the Routers

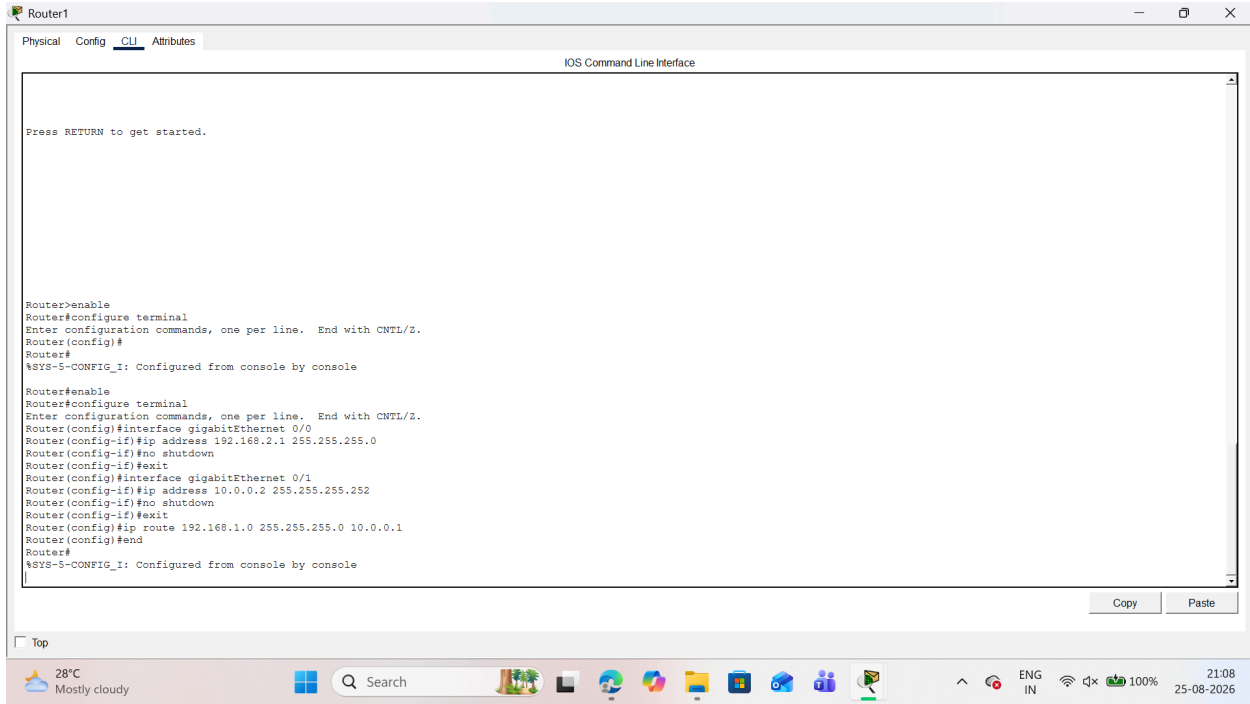
a.Router 0



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b.Router 1



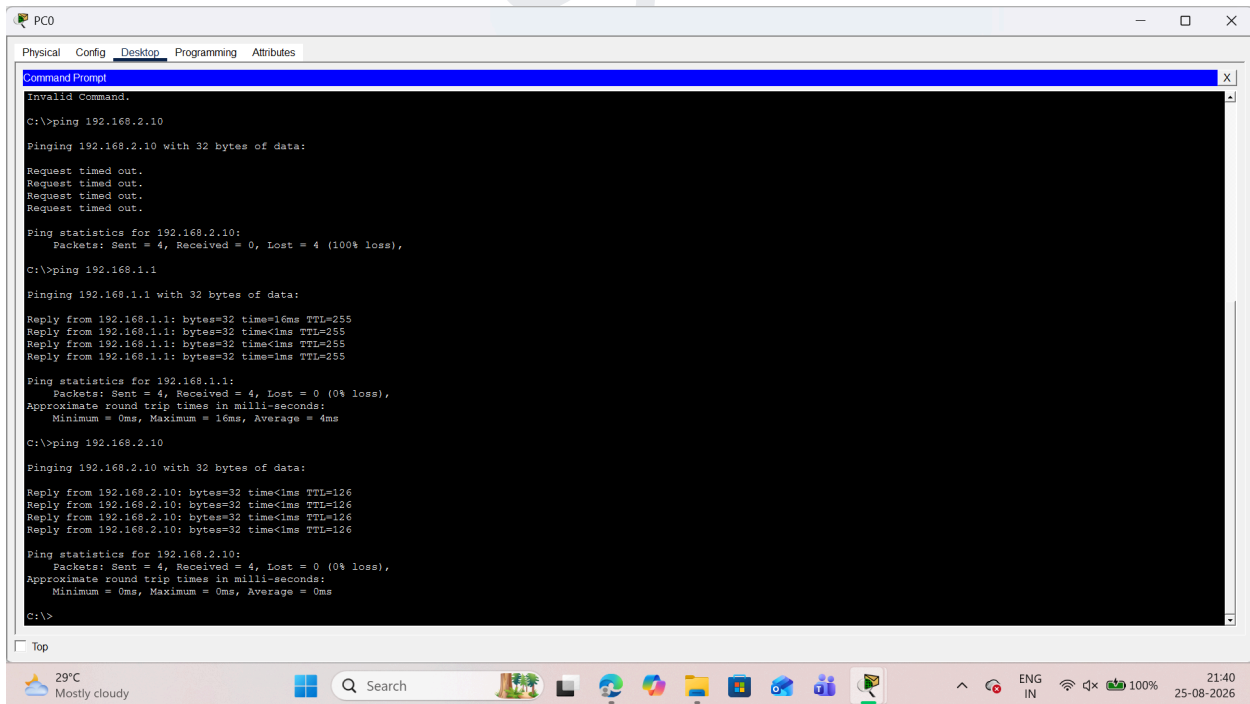
```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4. First test connectivity using ping command



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Invalid Command.

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=16ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 4ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

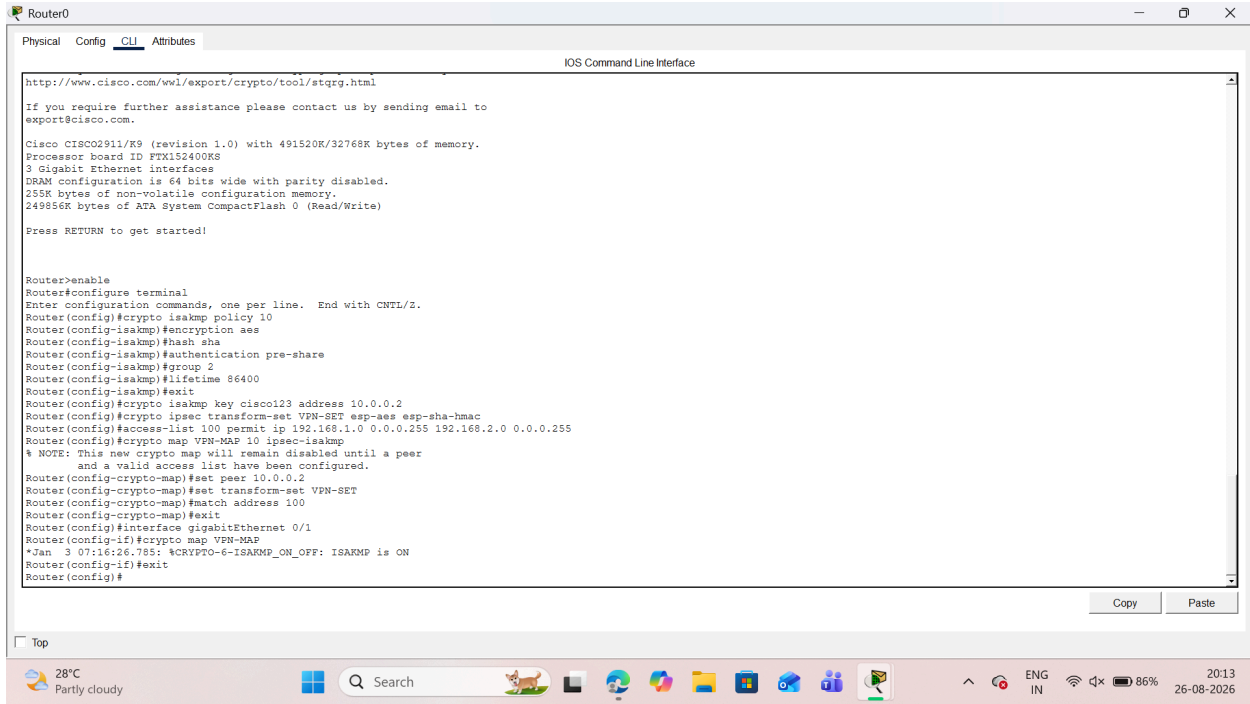
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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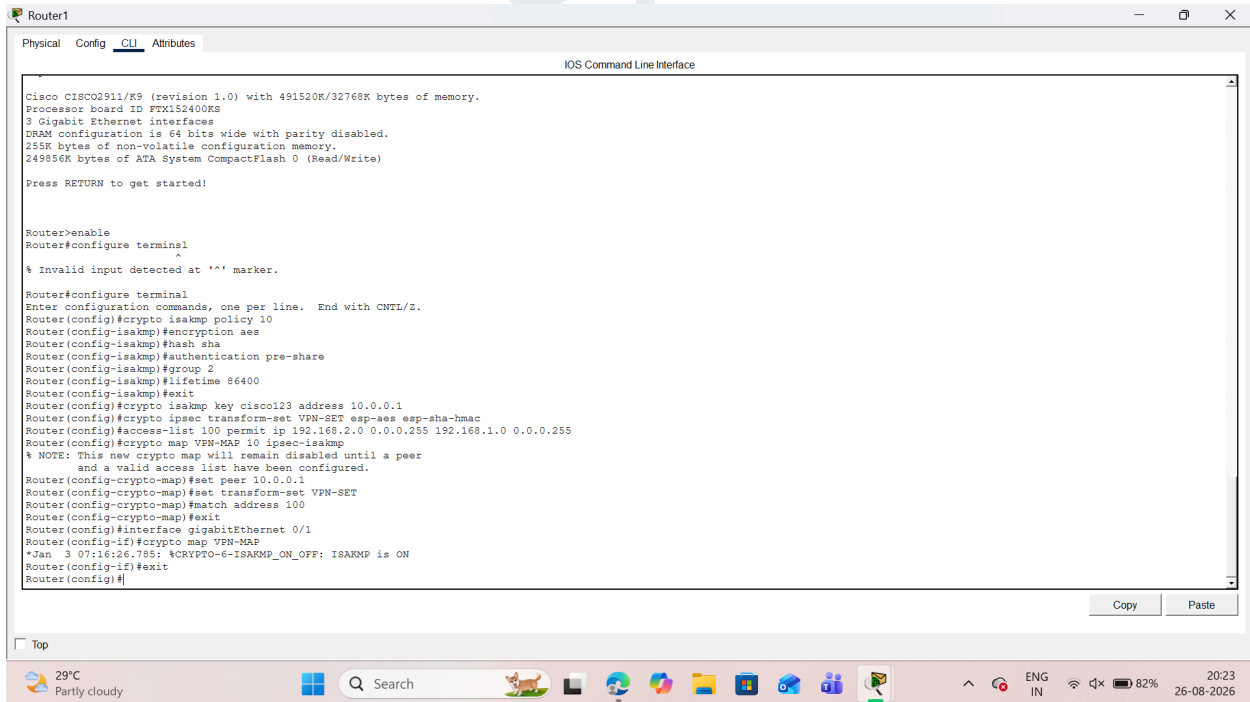
Step 5 . Configure IPsec on Router 0



The screenshot shows the CLI of Router0. The configuration includes enabling the router, setting terminal parameters, configuring an ISAKMP policy with SHA authentication and a pre-shared key, creating an IPsec transform set with ESP-AES and SHA-HMAC, and applying it to a crypto map on the GigabitEthernet0/1 interface. The configuration is successful, and the router status is shown as 'ON'.

```
Router0>enable
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router0(config)#crypto isakmp policy 10
Router0(config-isakmp)#encryption aes
Router0(config-isakmp)#hash sha
Router0(config-isakmp)#authentication pre-share
Router0(config-isakmp)#group 2
Router0(config-isakmp)#lifetime 86400
Router0(config-isakmp)#exit
Router0(config)#crypto isakmp key cisco123 address 10.0.0.2
Router0(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router0(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Router0(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router0(config-crypto-map)#set peer 10.0.0.2
Router0(config-crypto-map)#set transform-set VPN-SET
Router0(config-crypto-map)#match address 100
Router0(config-crypto-map)#exit
Router0(config)#interface gigabitEthernet 0/1
Router0(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router0(config-if)#exit
Router0(config)#
```

Step 6 Configure IPsec on Router 1

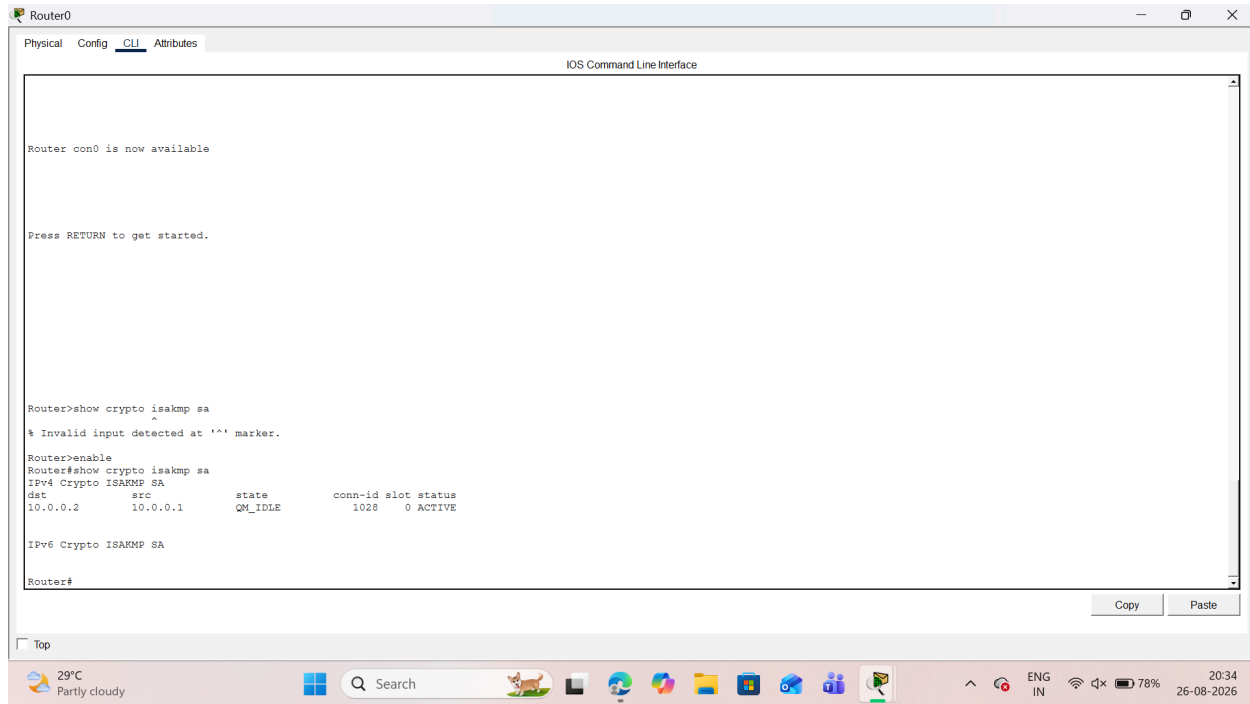


The screenshot shows the CLI of Router1. The configuration is identical to Router0, but it includes an error message: '% Invalid input detected at '^' marker.' This occurs because the configuration was pasted from Router0's output, which includes a carriage return character. The configuration is otherwise successful, and the router status is shown as 'ON'.

```
Router1>enable
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#crypto isakmp policy 10
Router1(config-isakmp)#encryption aes
Router1(config-isakmp)#hash sha
Router1(config-isakmp)#authentication pre-share
Router1(config-isakmp)#group 2
Router1(config-isakmp)#lifetime 86400
Router1(config-isakmp)#exit
Router1(config)#crypto isakmp key cisco123 address 10.0.0.1
Router1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router1(config)#access-list 100 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
Router1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router1(config-crypto-map)#set peer 10.0.0.1
Router1(config-crypto-map)#set transform-set VPN-SET
Router1(config-crypto-map)#match address 100
Router1(config-crypto-map)#exit
Router1(config)#interface gigabitEthernet 0/1
Router1(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Router1(config-if)#exit
Router1(config)#
```

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Step 7 Test the IPsec tunnel and verifying



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router con0 is now available

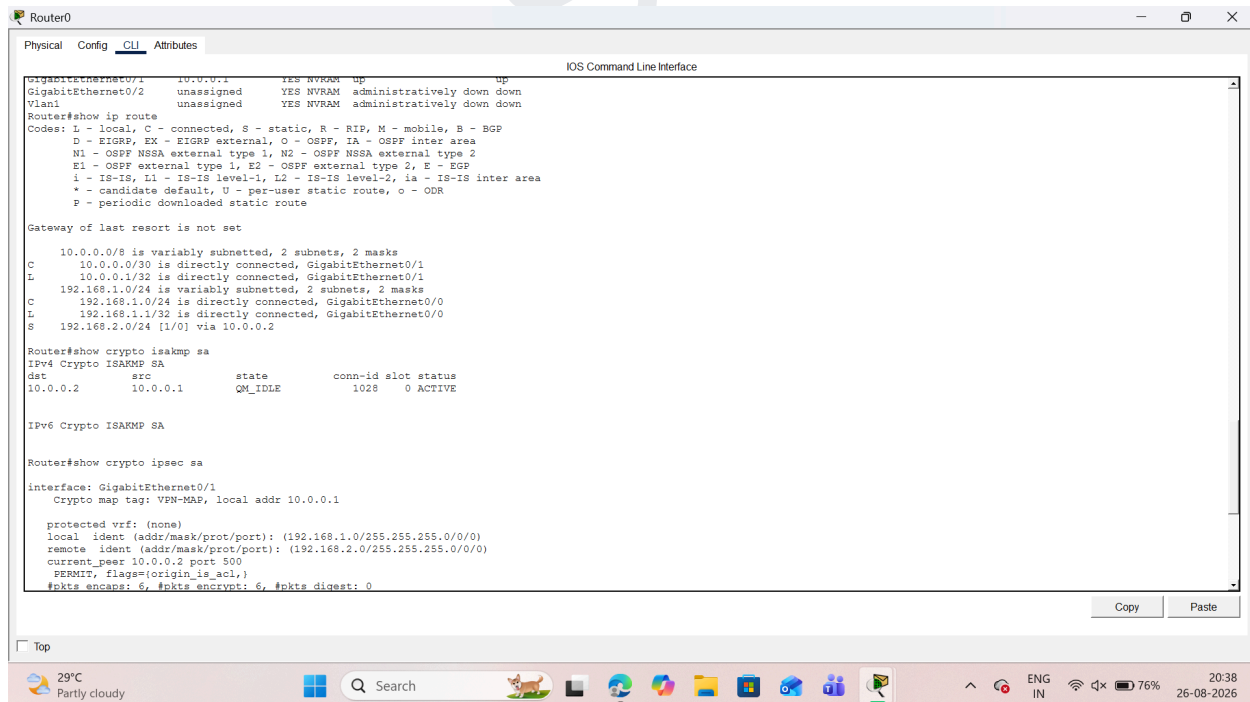
Press RETURN to get started.

Router>show crypto isakmp sa
^
% Invalid input detected at '^' marker.

Router>enable
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state      conn-id slot status
10.0.0.2  10.0.0.1  QM_IDLE    1028     0  ACTIVE

IPv6 Crypto ISAKMP SA

Router#
```



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

GigabitEthernet0/1  10.0.0.1  YES NVRAM  up      up
GigabitEthernet0/2  unassigned YES NVRAM  administratively down down
Vlan1               unassigned YES NVRAM  administratively down down

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
  C    10.0.0.0/30 is directly connected, GigabitEthernet0/1
  L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
  L    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
  C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
  L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
  S    192.168.2.0/24 [1/0] via 10.0.0.2

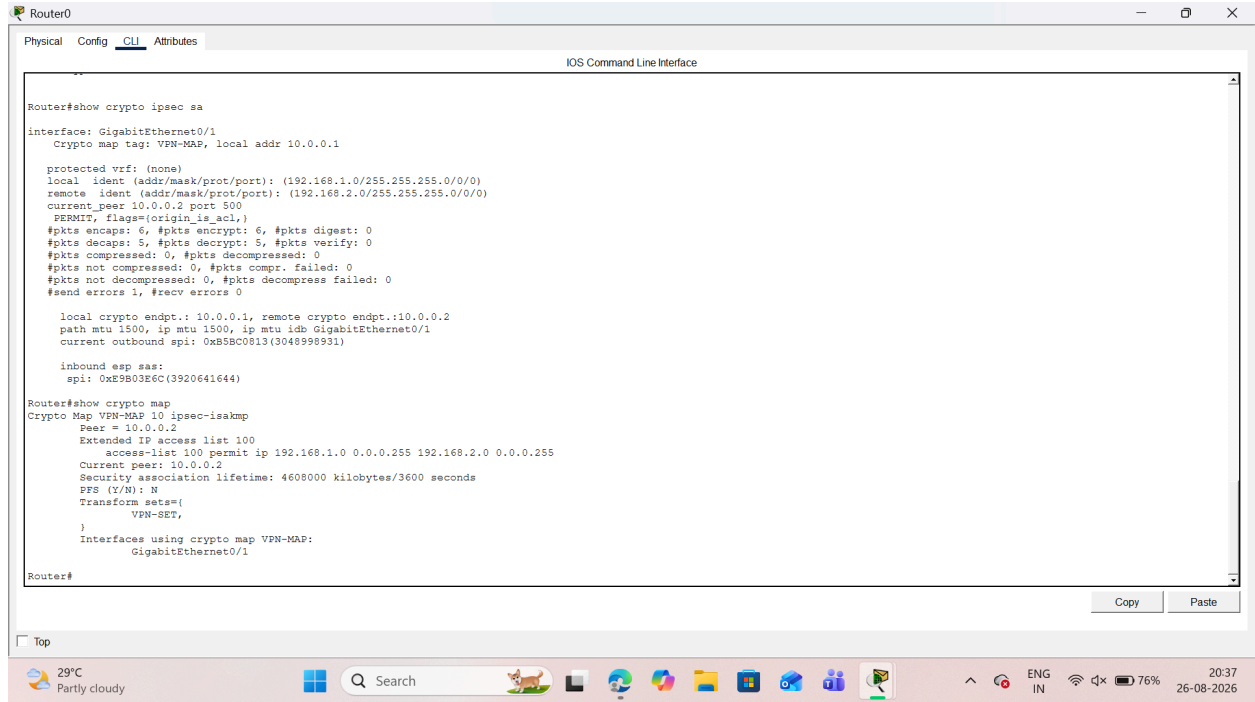
Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst      src      state      conn-id slot status
10.0.0.2  10.0.0.1  QM_IDLE    1028     0  ACTIVE

IPv6 Crypto ISAKMP SA

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
  Crypto map tag: VPN-MAP, local addr 10.0.0.1

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.1.0/255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.2.0/255.255.0/0/0)
  current_peer 10.0.0.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
```

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```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#show crypto ipsec sa
interface: GigabitEthernet0/1
Crypto map tag: VPN-MAP, local addr 10.0.0.1

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.2.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
PERMIT, flags=(origin_is_acl,)
#pkts encaps: 6, #pkts encrypt: 6, #pkts digest: 0
#pkts decaps: 5, #pkts decrypt: 5, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

local crypto endpt.: 10.0.0.1, remote crypto endpt.: 10.0.0.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0x5B8C0813(3048998931)

inbound esp sas:
spi: 0xE9B03E6C(3920641644)

Router#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
Peer = 10.0.0.2
Extended IP access list 100
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
Current peer: 10.0.0.2
Security association lifetime: 4608000 kilobytes/3600 seconds
PFS (Z/N): N
Transform sets=
VPN-SET,
Interfaces using crypto map VPN-MAP:
GigabitEthernet0/1

Router#
```

Top

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